Power equations

1

a $x = 3$ **b** $x = 8$ **c** $x = \frac{8}{5}$ **d** $x = 6$

e $x = 2$ **f** $x = 6$ **g** $x = 5$ **h** $x = 2$

Exponential equations

2

a $x = 8$ **b** $x = -4$ **c** $x = \frac{2}{9}$ **d** $x = 3$

e $x = 0$ **f** $x = -2$ **g** $x = 7$ **h** $x = -2$

i $x = -3$ **j** $x = \frac{1}{3}$ **k** $x = \frac{1}{2}$ **l** $n = \frac{1}{3}$

3

a $3 < x < 4$ **b** $3 < x < 4$

c $-4 < x < -3$ **d** There is no real solution.

4

a **i** $2^{3x} = 2^2$ **ii** $x = \frac{2}{3}$

b **i** $2^{4x} = 2^{-1}$ **ii** $x = -\frac{1}{4}$

c **i** $2^{-10} = 2^{2x}$ **ii** $x = -5$

d **i** $2^{\frac{x}{2}} = 2$ **ii** $x = 2$

5

a $y = \frac{3}{2}$ **b** $x = 6$ **c** $y = 4$ **d** $k = -2$

e $x = 2$ **f** $x = -2$ **g** $x = -3$ **h** $x = \frac{7}{3}$

i $x = -7$ **j** $x = \frac{21}{19}$ **k** $y = \frac{11}{6}$ **l** $x = -\frac{41}{6}$

m $x = 5$ **n** $x = 1$ **o** $x = -\frac{11}{2}$ **p** $x = \frac{21}{4}$

q $x = -7$ **r** $x = 2$ **s** $x = 5$ **t** $x = \frac{5}{2}$

u $x = 4, x = -1$ **v** $x = 3$

6

a **i** 3^{x+nx} **ii** $x = \frac{4}{1+n}$

b **i** 3^{3x-2k} **ii** $x = 1 + \frac{2k}{3}$

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7

a
- i $m = 2^x$
- ii $x = 3,\ 0$

b
- i $m = 2^x$
- ii $x = 3,\ 2$

c
- i $m = 2^x$
- ii $x = -1,\ 3$

d
- i $m = 4^x$
- ii $x = 0,\ 3$

e
- i $m = 2^x$
- ii $x = 0,\ 2$

f
- i $m = 3^x$
- ii $x = 1,\ 2$

8 8

Applications

9 $x = -2$

10 $h = 2$

11

a $n = 8$ **b** $k = 4$ **c** $m = -2$

12

a $t = 5$ hours **b** $t = 10$ hours **c** $t = 12$ hours

13

a 440 Hz **b** 262 Hz **c** $n = 73$

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